AEO “Nazarbayev Intellectual Schools”

CHEMISTRY **Grade 12**

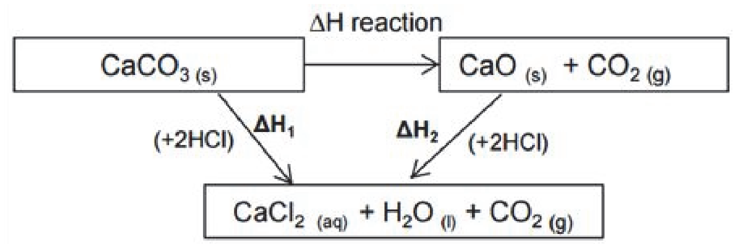
Paper 3 **June 2024**

MARK SCHEME

Maximum Mark: 30

Question	Answer	Mark	Additional Guidance
1(a) (i)	$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$	[1]	
1(a) (ii)	Reducing agent – Fe^{2+} Oxidising agent - MnO_4^-	[2]	
1(b) (i)	for completing table titrations used – 2 and 3 mean titre – 27.00 / 26,95	[3]	**accept** mean quoted to 1, 2 or 3 decimal places correctly rounded
1(b) (ii)	Moles of MnO_4^- = $0.0200 \times \frac{27}{1000} = 5.4 \times 10^{-4}$	[1]	**ecf** from **1(b)(i)**
1(b) (iii)	Moles in 25 cm^3= $5.4 \times 10^{-4} \times 5 = 2.7 \times 10^{-3}$/0.0027 Moles in 250 cm^3= 2.7×10^{-2}/ 0.027	[2]	**Accept** correct answer award 2 marks **ecf** from **1(b)(ii)**
1(b) (iv)	(mass of Fe^{2+}) = (moles of Fe^{2+}) × 56 =0.027× 56 =1.51g	[1]	**ecf** from **1(b)(iii)**
1(b) (v)	percentage by mass = $\frac{1.51}{4.50} \times 100\% = 33.6\%$ /	[1]	**Accept** 33.55% **ecf** from **1(b)(iv)**
1(c)(i)	Student was incorrect. Equivalent point occurs before end point, colour change occurs at the end point.	[1]	**owtte**
1(c)(ii)	percentage error = $\frac{0.1}{27.00} \times 100\% = 0.37\%$	[1]	**Accept** 26,95 instead of 27 in calculation
1(c)(iii)	pipette	[1]	**Accept** burette/ electronic dropper

Question	Answer	Mark	Additional Guidance
2(a)	The enthalpy (energy) change for converting reactants into products is the same regardless of the route taken	[1]	Idea of the total enthalpy change of a reaction is independent of the reaction route
2 (b) (i)	3.5 ^0C	[1]	
2 (b) (ii)	735J	[1]	**Ignore** sign
2 (b) (iii)	0.025 mole - 29.4 kJ mol^{-1}	[2]	**Do not accept** 29.4
2 (c) (i)	correct labels **and** suitable linear scales for both axes all points plotted to within half a square appropriate lines of best fit drawn. AND either a straight line / smooth curve after the max T Lines extrapolated and correct value of $\Delta T = 15{,}5 \pm 2$	1 1 1 1 [max 4]	**ecf** from straight lines on the plotted points
2 (c) (ii)	Anomalous points are marked. Not consequently stirring the mixture Or Wrong reading of thermometer scale Or	[1]	**Owtte** **Do not accept** Human error

	Change the conditions of experiment		
2 (c) (iii)	*Calcium carbonate results* q = heat released = 50 x 4.2 x temperature increase (J)	[1]	
2 (c) (iv)	enthalpy changes calculated = (-q ÷ 0.025)/1000	[1]	answer must have negative sign not answer with +
2 (d)(i)	**Attempt at Hess cycle diagram** • downward arrow from $CaCO_3$, labelled with value and ΔH_1 from (b)(i) • downward arrow from CaO, labelled with value and ΔH_2 from (b)(ii)	[2]	 **ignore** state symbols
2 (d)(ii)	ΔH_r *value* = ($\Delta H_{experiment\ 1} - \Delta H_{experiment\ 2}$)	[1]	sign must be correct from the enthalpy changes, but ignore missing + if the answer comes out positive
2 (e)	Acid in excess (in both cases) Therefore, temperature rise would be same	[1]	**accept** CaO is limiting reagent